Signal Processing 198 (2022) 108561

Contents lists available at ScienceDirect

Signal Processing

journal homepage: www.elsevier.com/locate/sigpro

Audio-lossless robust watermarking against desynchronization attacks

Jingxuan Li, Shijun Xiang*

College of Information Science and Technology/College of Cyber Security, Jinan University, Guangzhou 510632, China

ARTICLE INFO

Article history:
Received 17 September 2021
Revised 28 March 2022
Accepted 29 March 2022
Available online 31 March 2022

Keywords:
Audio-lossless robust watermarking
Discrete wavelet transform
Histogram shape
Desynchronization attacks
Compensation information

ABSTRACT

Audio-lossless robust watermarking (ALRW) is a new research issue in the field of information hiding, which can resist those content-preserving manipulations while can restore the cover audio files completely in case of no attacks. Most ALRW countermeasures proposed in the literature usually focus on additive noise-like manipulations such as MP3 compression, low-pass filtering and Gaussian additive noise, but few are resistant to challenging desynchronization attacks such as random cropping and time-scale modification (TSM). The main reason is that the exploited robust features in the existing ALRW algorithms are related to the sample position. In this paper, we propose an ALRW method by efficiently embedding a watermark into the histogram shape of the low-frequency component after the discrete wavelet transform (DWT) and reversibly hiding the distortion due to the robust watermark embedding as the compensation information for restoration of the cover audio file. The use of the histogram shape in the DWT low-frequency component as the exploited robust feature is: 1) mathematically invariant to TSM; 2) robust to random cropping attacks since an audio clip before and after the cropping often has similar distribution, and 3) resistant to interpolation errors during time scaling and those common audio processing operations. To reduce the compensation information, the robust watermarking process is elaborately and luminously designed to represent the difference between the original and the robust watermarked audio. As a result, an ALRW system against desynchronization attacks is achieved with satisfactory performance. Experimental results show that the proposed ALRW scheme can effectively reduce the compensation information and provide strong robustness to those common manipulations including TSM, random cropping, MP3 lossy compression and other noise-like manipulations. In case of no attacks, the cover audio signal can be recovered without any loss.

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1. Introduction

With the rapid development of computer and Internet technology, personal computers and Internet technology have already entered thousands of households. The development of computers has had a profound impact on people's daily lives. People can use computer technology to realize the ancient desire to know everything about the world without leaving home. However, the development of science and technology is a double-edged sword. While bringing convenience to people, it also provides convenience to some criminals. Once the criminals intercept the digital multimedia file in the process of transmission, they can illegally tamper with the content of the intercepted file or destroy the copyright information. Therefore, how to protect digital multimedia files from illegal infringement has become an important research direction for researchers all over the world. To solve this problem, researchers have proposed many various solutions, such as digital watermarking technology. According to different application scenarios of multimedia files, digital watermarking can be divided into fragile watermarking [1–3] and robust watermarking (RoW) [4–6]. In most situations, fragile watermarking is for integrity authentication, while robust watermarking is for copyright protection.

As one kind of fragile watermarking, reversible watermarking (ReW) for audio means that the receiver can extract watermark information from the watermarked audio file without any distortion and restore the original audio file in a lossless way [7]. The hash value of the cover file is usually part of the payload for authentication. The reversible watermark information can be embedded in the time domain [8–10] and the transform domain [11–13] for different purposes. In the time domain, Yan and Wang [8] proposed a new ReW algorithm by using the prediction error expansion technique, which has a high watermark embedding capacity while better maintaining the imperceptibility of the watermarked audio. In [9], Bobeica *et al.* proposed a capacity control algorithm for audio watermarking, which can provide superior distortion/bitrate results. Inspired by the image ReW algorithm,

* Corresponding author.:.
E-mail address: shijun_xiang@qq.com (S. Xiang).

https://doi.org/10.1016/j.sigpro.2022.108561
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Bobeica *et al.* [10] proposed an audio reversible data hiding algorithm based on audio sample value ordering. In the transform domain, Choi and Pun [11] proposed a high-capacity audio ReW algorithm based on generalized integer transform, which uses the difference expansion technique to embed the reversible watermark information. In [12], Naskar *et al.* combined the quad transform and the redundant embedding technique to embed the reversible watermark. Beyond that, in [13], Ginanjar *et al.* proposed a novel audio watermarking algorithm for high-quality audio files by using the fast Fourier transform (FFT) and traditional phase shift keying (PSK) techniques to embed the watermark information into the digital audio signal. The ReW has the superiority of recovering the cover file but is sensitive to any attacks, even modification of a bit in the marked audio file.

As for robust audio watermarking, in recent years, many robust audio watermarking algorithms have been developed [14], which can be divided into two categories: additive noise-like manipulations and desynchronization attacks. On the one hand, for additive noise-like manipulations, Xiang *et al.* [15] proposed a new audio watermarking algorithm based on spread spectrum for satisfactory robustness to lossy compression. Fallahpour and Megias [16] presented a high-capacity audio watermarking algorithm using the FFT spectrum, which has excellent robustness to those common operations. Xiang *et al.* [17] proposed a novel robust audio watermarking algorithm based on spread spectrum for severe noise addition and compression attacks. Furthermore, in [18], Wang *et al.* presented a multi-subspace echo information hiding algorithm, which achieves higher robustness and security. On the other hand, desynchronization attacks include time-scaling modification (TSM), cropping, and jitter attacks, which are more difficult to resist in robust audio watermarking community. Under a desynchronization attack, the watermark is still in the marked file, but it can not be detected due to the modification of sample positions. For desynchronization attacks, Li and Xue [19] proposed an algorithm based on audio content analysis, which has good robustness to synchronization attacks by using the peak points as a reference. In [20–27], watermark robustness to desynchronization attacks was achieved by using the invariant characteristics of histogram shape to sample positions. In [28–38], some robust features to desynchronization attacks were designed for inserting a robust watermark. Although all the above robust audio watermarking algorithms have strong robustness, the original audio cannot be recovered after the robust watermark is extracted.

On the basis of ReW and RoW techniques, some researchers have proposed several robust reversible watermarking algorithms with the properties of both robustness and reversibility. Considering that these methods are robust and reversible, we call them cover-lossless robust watermarking techniques in this paper, such as image-lossless robust watermarking methods, audio-lossless robust watermarking (ALRW) schemes, etc. In other words, a cover-lossless robust watermarking algorithm means that when the multimedia file is not attacked during transmission, the receiver can extract the robust watermark and restore the original file completely. At the same time, if the watermarked multimedia file is attacked, the receiver can still extract the robust watermark for copyright protection. At present, the cover-lossless robust watermarking schemes can be divided into two categories: histogram shifting based methods and two stages of watermarking schemes.

- **For histogram shifting based methods:** In [39], Zeng *et al.* proposed an image-lossless robust watermarking scheme using the histogram shifting algorithm, which has the robustness to JPEG compression attacks. An *et al.* [40] constructed a novel watermark embedding and extraction algorithm by using wavelet-domain statistical quantity histogram shifting and clustering (WSQH-SC). Xiang *et al.* [41] introduced a novel ALRW algorithm for lossy compression. For a numerical relational database, Hu *et al.* [42] combined the genetic algorithm and histogram shifting method to present a database-lossless robust watermarking technique. Furthermore, an ALRW algorithm based on high-order difference statistics was proposed in [43]. Although these histogram-based approaches can resist additive noise-like manipulations, such as MP3 compression, noise, filtering, they cannot resist geometric attacks and desynchronization attacks because the exploited robust features are relevant to the pixels or sample positions.
- **For two stages of watermarking schemes:** Coltuc and Chassery [44] presented a two-stage image-lossless robust watermarking framework by luminously combining RoW and ReW techniques, which is one of the multiple watermarking techniques. This framework can provide satisfactory robustness and restore the original cover without any distortion in case of no attacks. The basic principle is to embed a robust watermark into an image at first and then insert the distortion due to the robust watermark embedding into the marked image in a lossless way to generate the final watermarked image. In case of no attacks, the distortion can be extracted for recovery of the original file. In the literature, researchers have designed various cover-lossless robust watermarking methods by using the kind of two stages of watermarking technique, e.g., for images [45–53] and audio files [54,55]. In these methods, few can resist geometric distortions or desynchronization attacks due to the fact that the robust watermark is sensitive to the spatial or time position. Recently, Hu and Xiang [52,53] proposed a two-stage watermarking scheme by embedding the robust watermark into invariant moments so as to achieve robustness for geometric deformations. In the field of digital audio, there are currently no any reports on ALRW for desynchronization attacks, such as TSM and cropping attacks.

In this paper, we proposed a two-stage ALRW algorithm, which can resist desynchronization attacks (e.g., TSM, cropping, jitter) well by embedding the robust watermark into the histogram shape of audio. On the one hand, if the watermarked audio is not attacked, the receiver can extract the robust watermark from the watermarked audio and recover the original audio without any distortion. On the other hand, if the watermarked audio is maliciously attacked in some way during transmission, the receiver can still extract the robust watermark from the attacked audio for copyright protection. Experimental results show that the proposed algorithm can resist both additive noise-like operations (e.g., filtering, MP3 compression, etc.) and desynchronization attacks (e.g., TSM, cropping, jitter, etc.) and recover the cover file under no attacks.

The main contribution of this paper is to propose an ALRW method, which can overcome the challenging desynchronization attacks. Compared with the existing audio watermarking algorithms, the highlights of this article are as follows,

- The proposed ALRW scheme in this article can resist both additive noise-like operations and desynchronization attacks, while the previous schemes are only robust to additive noise-like operations.
- In case of no attacks, the traditional robust audio watermarking algorithms cannot restore the original audio, while the ALRW method proposed in this paper can recover the original audio in a lossless way after extracting the robust watermark.
- The proposed ALRW scheme can be used in more security applications for integrity authentication (by its reversibility) and copyright protection (by its excellent robustness).

In the next section, we briefly introduce the desynchronization attacks, previous algorithms based on the invariance of histogram shape, and related works. Then we elaborate on the ALRW algorithm proposed in this paper. Immediately afterward, we analyze

and explain the experimental results. Finally, we make a summary and look forward to future works.

2. Preliminary

In this section, desynchronization attacks and some related works on the use of the histogram shape will be briefly introduced.

2.1. Desynchronization attacks

In the field of digital audio watermarking, desynchronization attacks are one of the most challenging attacks. Desynchronization attacks in the field of robust audio watermarking mainly include TSM, cropping, and jitter. Specifically, the TSM attack includes pitch invariance TSM and resample TSM. Pitch invariance TSM can preserve the pitch of the audio signal. Resample TSM keeps the audio pitch and tempo neither by changing playback speed. Random cropping refers to randomly cutting out a part or several pieces from the original audio. Similar to random cropping, jitter attack refers to deleting a part of audio at every interval in the time domain. All of these desynchronization attacks will cause greater damage to the structure of original audio media. As a result, when watermarked audio is subjected to these attacks, the synchronization structure of the watermarked audio will be destroyed, making it difficult for the receiver to extract the robust watermark from the attacked audio.

2.2. The use of the invariance of histogram shape

In previous works, various RoW algorithms using the invariance of the histogram shape to resist geometric attacks or desynchronization attacks have been proposed in [20–27,56]. The basic strategy to use the invariance of the histogram shape is to divide a few bins as a group to embed bits, each group for one watermark bit:

- **Two bins as a group**: The methods in [24,56] divided every two adjacent bins in the histogram into a set to embed one-bit robust watermark information. To be specific, in [56], Xiang *et al.* embedded robust watermark information into the image by using the invariance of the histogram shape in the low-frequency component of the Gaussian low-pass filtering. Later, in [24], the robust watermark information was inserted into the audio files.
- **Three bins as a group**: In [20–22,25,27], three adjacent bins in the histogram were divided into a group to embed one-bit robust watermark information. In [20], Xiang *et al.* proved the invariance of the histogram shape to time scaling in the time domain and then proposed a robust audio watermarking algorithm that can resist various desynchronization attacks including TSM, cropping, and jitter attacks. In [22], Xiang *et al.* theoretically proved that the invariance of the histogram shape in the time domain can be transferred to the low-frequency subband of the discrete wavelet transform (DWT) for robustness improvement to additive noises. Furthermore, in [27], Hua *et al.* proposed two informed histogram-based audio watermarking schemes to improve the imperceptibility and robustness of watermarking by using the histogram shape.
- **Four bins as a group**: Moreover, in [23,26], the authors exploited every four adjacent bins in the histogram into a group to embed one bit. Specifically, Zhang *et al.* [23] proposed a robust audio watermarking scheme, which exploits the histogram shape to embed the robust watermark information. Yang *et al.* [26] presented a robust audio watermarking algorithm by combining the undecimated DWT and the invariance of the histogram shape to time scaling.

The use of two bins as a group to embed one bit is beneficial to enhancing the capacity. The more bins used as a group for embedding one bit is beneficial to improving the robustness of the watermark, but due to the limitation of the bin number with a fixed bin width, the embedding capacity will be reduced. In practical applications, a trade-off between robustness and capacity is necessary.

We can see from the above description that a watermark can be embedded into the histogram of the low-frequency component of the Gaussian filter or in the low-frequency component after performing an orthogonal transform for both additional noises and desynchronization attacks. Next, we will introduce the proposed ALRW algorithm by elaborately considering the capacity of the robust watermark, the amount of additional information, and the imperceptivity of the marked clips. Under the constraint of robustness, 1) two consecutive bins are designed as a group for inserting robust bits as much as possible; 2) for the reversibility and the imperceptivity, the additional information to be embedded reversibly for recovering the original audio should be as small as possible. From these two angles, we proposed an ALRW strategy to embed the robust watermark into the DWT low-frequency component combined with an efficient ReW way. The detail is as follows.

3. Proposed audio-lossless robust watermarking algorithm

In this section, we introduce the proposed ALRW algorithm, which can resist those common additive noise-like operations and desynchronization attacks. The invariance of the histogram shape in the low-frequency sub-band of DWT is introduced to embed a robust message. After the robust watermark is embedded, the distortion due to the robust watermark embedding is represented as the compensation information and embedded into the robust watermarked audio for recovering the original audio at the receiver side. The proposed ALRW algorithm organically combines an efficient RoW algorithm and a high-capacity ReW algorithm.

Fig. 1 illustrates the proposed ALRW algorithm, which includes several main blocks: robust watermark embedding, computing and inserting the additional information reversibly, extraction of additional information, integrity authentication, extraction of the robust watermark, and recovery of the cover audio. On the embedding side, the robust watermark bits are first embedded in the low-frequency component of DWT to generate the robust watermarked audio F_{w1}. Then, the additional information including the compensation information C and hash value H of F_{w1} is calculated and embedded into F_{w1} by using a high-capacity ReW algorithm. Finally, the watermarked audio F_w is obtained. At the receiving end, the compensation information C_1 and hash value H_1 are extracted at first. Then, we obtain the robust watermarked audio F'_{w1} and compute the hash value of F'_{w1}, denoted by H_2. If $H_1 = H_2$, the watermarked audio is not attacked during transmission. In this case, the robust watermark sequence W can be extracted, and the original audio F can be recovered in a lossless way. On the contrary, if $H_1 \neq H_2$, the watermarked audio has been attacked during the transmission process. In this case, the robust watermark sequence W' can be directly extracted from the received audio for copyright authentication.

3.1. Preprocessing

Suppose that there is a binary sequence (e.g., a serial number, ID number, etc.) $W = \{w(i)|i = 1, \ldots, L_w\}$ as the robust watermark to be embedded in an audio sequence $F = \{f(i)|i = 1, \ldots, N\}$. In order to improve the security of the algorithm proposed in this paper, the robust watermark W is encrypted at first by using the logistic chaotic sequence introduced in [38,57] before embedding.

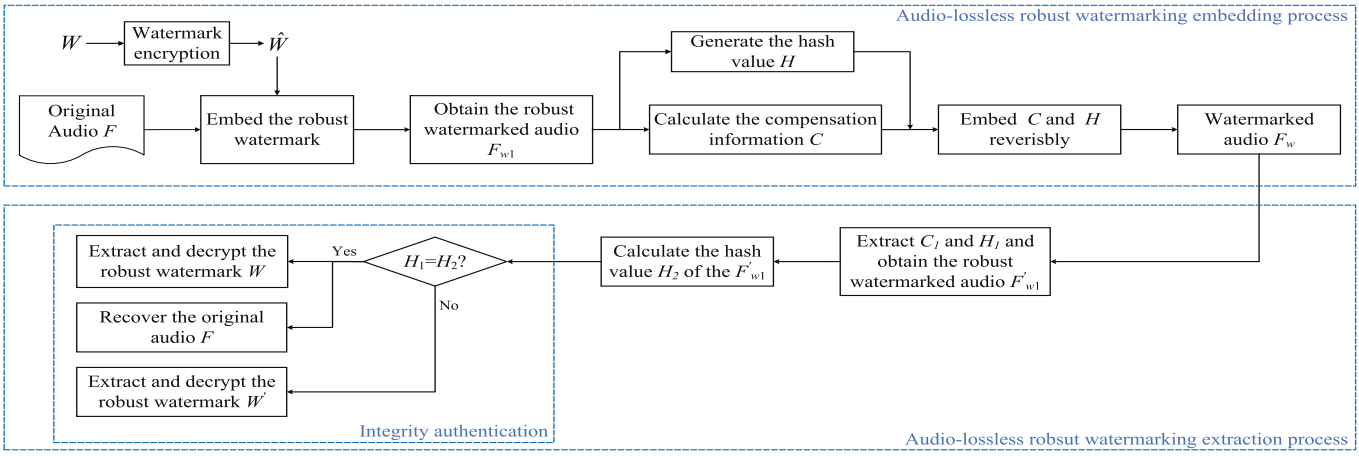


Fig. 1. The proposed ALRW framework.

Suppose that the logistic chaotic sequence R is generated as

$$R(i) = p \times R(i-1)[1 - R(i-1)] \tag{1}$$

where $R(i)$ is the ith element in R, $i = 2, 3, \ldots, L_w$, $R(1)$ is a random number of (0,1), and $p \in (3.5699456, 4)$. Then, the encrypted sequence EN is calculated by

$$EN(i) = \begin{cases} 0, & if\ R(i) < k \\ 1, & else \end{cases} \tag{2}$$

where $EN(i)$ is the ith element in EN, $i = 1, 2, \ldots, L_w$, and k is a number within (0,1). The robust watermark sequence $\hat{W} = \{\hat{w}(i) | i = 1, 2, \ldots, L_w\}$ after performing the logistic chaotic encryption can be obtained as

$$\hat{w}(i) = w(i) \oplus EN(i) \tag{3}$$

where $\oplus$ denotes XOR operation. In this paper, $R(1)$ is 0.78, p is 3.5869352, and k is 0.76. The sequence EN is used as a secret key to guarantee the security of the watermark in this article. Immediately after, the mean value of the original file, denoted by A, is calculated by

$$A = \frac{1}{N} \sum_{i=1}^{N} |f(i)| \tag{4}$$

Next, we select the samples in the amplitude range $B = [-\lambda A, \lambda A]$ to calculate the time-domain histogram $HS = \{hs(i) | i = 1, ., L\}$, where $L \geq 2L_w$ and λ is a positive number. Then, in each bin each group of each four continuous samples in position will be organized to perform a two-layer Haar DWT to obtain the low-frequency histogram HS_1. Details on the Haar DWT used in this paper are described as follows.

3.2. Haar wavelet transform

Since the method proposed in this paper needs to recover the original audio without any distortion after the robust watermark is extracted (in case of no attacks), an integer DWT operation is required for embedding the watermark. In this paper, the introduced Haar DWT can separate a digital audio signal into the low-frequency and high-frequency components in which the coefficients are integers. Because of that, it is beneficial to avoid the computation errors in the watermarking system and keep the reversibility. Assuming that (s_1, s_2) are two adjacent amplitudes in digital audio, the Haar DWT on (s_1, s_2) is calculated as

$$\begin{cases} s^l &= \lfloor \frac{s_1 + s_2}{2} \rfloor \\ s^h &= s_1 - s_2 \end{cases} \tag{5}$$

where $\lfloor \cdot \rfloor$ is the floor function, s^l denotes the low-frequency sub-band coefficient while s^h represents the high-frequency sub-band coefficient. Since the Haar DWT is orthogonal and invertible, the amplitude of the original audio signal (s_1, s_2) can be recovered without distortion by performing the inverse Haar wavelet transformation on s^l and s^h. The inverse Haar wavelet transformation process is computed as

$$\begin{cases} s_1 &= s^l + \lfloor \frac{s^h + 1}{2} \rfloor \\ s_2 &= s^l - \lfloor \frac{s^h}{2} \rfloor \end{cases} \tag{6}$$

Due to the fact that the Haar DWT is used, the audio signals before and after the DWT operation are all integers. As a result, there is no any distortion in the transformation operations and the robust distortion can be reversibly embedded into the robust watermarked audio. In this article, a two-layer Haar DWT is exploited to extract the low-frequency components.

3.3. Robust watermark embedding

After performing the two-layer Haar DWT operation, we calculate the statistics histogram of the low-frequency components, denoted by HS_1. Suppose two non-overlapping consecutive bins in the histogram HS_1 are divided into a group, denoted by $bin1$ and $bin2$, respectively. The numbers of the samples in $bin1$ and $bin2$ are a and b, respectively. We can embed one-bit robust watermark information by modifying a and b while keeping $a + b$ unchanged. The specific robust watermark embedding rule is

$$\begin{cases} \frac{a}{b} \geq T, & if\ \hat{w}(i) = 1 \\ \frac{b}{a} \geq T, & else \end{cases} \tag{7}$$

where T $(T \geq 1)$ is a threshold value. The larger T is, the stronger the robustness and the worse the imperceptibility; conversely, the smaller T is, the weaker the robustness and the better the imperceptibility. Therefore, the selection of T is a compromise between robustness and imperceptibility. If $\hat{w}(i) = 1$ and $\frac{a}{b} \geq T$, no operation is required. If $\hat{w}(i) = 1$ but $\frac{a}{b} < T$, I_1 samples will be selected from $bin2$, and the values of these selected samples will be modified to make them fall into $bin1$, satisfying $\frac{a'}{b'} \geq T$, where a' and b' are the numbers of samples in $bin1$ and $bin2$ after modification, respectively. The specific modification method is

$$f_2'(i) = f_2(i) - 2M, \quad 1 \leq i \leq I_1 \tag{8}$$

where $f_2(i)$ is the ith coefficient in $bin2$, $f_2'(i)$ is the modified coefficient, and $2M$ is the bin width in the low-frequency domain while

M is that in the time domain. I_1 is computed as

$$I_1 = \frac{Tb - a}{1 + T} \tag{9}$$

Similarly, if $\hat{w}(i) = 0$ and $\frac{b}{a} \geq T$, then no operation is required. If $\hat{w}(i) = 0$ but $\frac{b}{a} < T$, I_2 samples will be selected from *bin*1, and the values of these selected samples in *bin*1 are modified to make them fall into *bin*2. After that, the numbers of the samples in *bin*1 and *bin*2 satisfy the condition $\frac{b'}{a'} \geq T$. The specific modification method is

$$f_1'(i) = f_1(i) + 2M, \quad 1 \leq i \leq I_2 \tag{10}$$

where $f_1(i)$ is the ith coefficient in *bin*1, and $f_1'(i)$ is the modified coefficient. I_2 is calculated as

$$I_2 = \frac{Ta - b}{1 + T} \tag{11}$$

The above processes are repeated until all the robust watermark bits are embedded. Next, an inverse DWT (IDWT) is performed to obtain the robust watermarked audio F_{w1}.

3.4. Computation of additional information

After embedding the robust watermark, we will calculate the hash value of the robust watermarked audio and the compensation information used to recover the original audio at the receiving end. In this paper, the additional information mainly includes the hash value and the compensation information. The compensation information consists of the bin width M, the length of the robust watermark L_w, the selected audio amplitude range B, the selected constant λ, and the robust distortion D. The robust distortion can be formulated as

$$D_1(i) = F(i) - F_{w1}(i), i = 1, \ldots, N \tag{12}$$

where $D_1(i)$ is the distortion of the ith sample, by computing the difference between the ith original sample $F(i)$ and the ith marked sample $F_{w1}(i)$. For better compression, we performed the following operation on D_1 to get D_2,

$$D_2(i) = \begin{cases} 0, & if\ D_1(i) = 0 \\ 1, & if\ D_1(i) \neq 0 \end{cases} \tag{13}$$

In D_2, each element is '0' or '1'. Owe to the use of the two-layer discrete wavelet transform, the four consecutive '1' in D_2 can be represented as one '1' to get D, which has smaller size than D_2.

The additional information including M, L_w, B, λ, D, and H is compressed by using an arithmetic coding operation. After that, the compressed additional information is permutated to get the encrypted additional information by using another encrypted sequence EN_1 (which can be generated by Eqs. (1) and (2), similarly). Immediately afterward, the encrypted additional information is embedded into the robust watermarked audio F_{w1} reversibly by the following ReW way. Finally, the final watermarked audio F_w is obtained.

3.5. Embedding of additional information

To recover the original audio and verify the integrity of the audio in the receiver, the encrypted additional information is needed to be embedded into the robust watermarked audio reversibly. In this paper, on the basis of the high-capacity ReW algorithm proposed in [58], we designed the following reversible data hiding way for audio by

step 1 Dividing the audio sequence into dot set and cross set, as shown in Fig. 2.

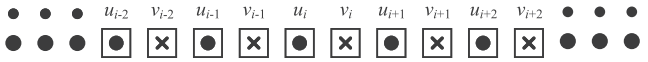


Fig. 2. Illustration on the dot and cross sets.

step 2 For the sample u_i in the dot set, we use the value v_{i-1} and v_i in the cross set as the context to get the predicted value x_i by

$$x_i = \lfloor (v_{i-1} + v_i)/2 \rfloor \tag{14}$$

step 3 Computing the prediction error e_i,

$$e_i = u_i - x_i \tag{15}$$

step 4 Embedding a bit m into e_i by using the following embedding rules,

$$E_i = \begin{cases} 2e_i + m, & if\ e_i \in [T_n; T_p] \\ e_i + T_p + 1, & if\ e_i > T_p\ and\ T_p \geq 0 \\ e_i + T_n, & if\ e_i < T_n\ and\ T_n < 0 \end{cases} \tag{16}$$

where T_p and T_n are positive threshold and negative threshold in the dot set, respectively.

step 5 Finally, the amplitude of the watermarked audio is computed by

$$u_i' = E_i + x_i \tag{17}$$

After the dot set has been embedded, we repeat the above *step2 - step5* to embed the remaining additional information on the cross set by using the marked dot set as the context.

3.6. Extraction of additional information

With the same ReW algorithm, on the receiver side, the encrypted additional information can be extracted as follows,

step 1 To divide the watermarked audio sequence into the dot set and cross set.

step 2 To extract the hidden message from the cross set at first by using the dot set as the context. The prediction value x_i' of the watermarked value v_i' is calculated by the following equation,

$$x_i' = \lfloor (u_i' + u_{i+1}')/2 \rfloor \tag{18}$$

where u_i' and u_{i+1}' are the two watermarked sample values in the dot set.

step 3 After that, the prediction error E_i' is obtained by

$$E_i' = v_i' - x_i' \tag{19}$$

step 4 After the prediction error is obtained, the reversible watermark m' in the cross set is extracted as following rules:

$$m' = E_i'\ mod\ 2,\ E_i' \in [2T_n', 2T_p' + 1] \tag{20}$$

where T_p' and T_n' are positive threshold and negative threshold in the cross set, respectively.

step 5 Next, the original prediction error e_i' in the cross set is calculated as follows:

$$e_i' = \begin{cases} \lfloor E_i'/2 \rfloor, & if\ E_i' \in [2T_n'; 2T_p' + 1] \\ E_i' - T_p' - 1, & if\ E_i' > 2T_p' + 1\ and\ T_p' \geq 0 \\ E_i' - T_n', & if\ E_i' < 2T_n'\ and\ T_n' < 0 \end{cases} \tag{21}$$

step 6 Finally, the amplitude value of the robust watermarked audio v_i in the cross set is recovered by

$$v_i = x_i' + e_i' \tag{22}$$

Similarly, after the cross set has been extracted, we repeat the above *step2 - step6* to extract the reversible watermark information in the dot set. Finally, we restore the robust watermarked audio.

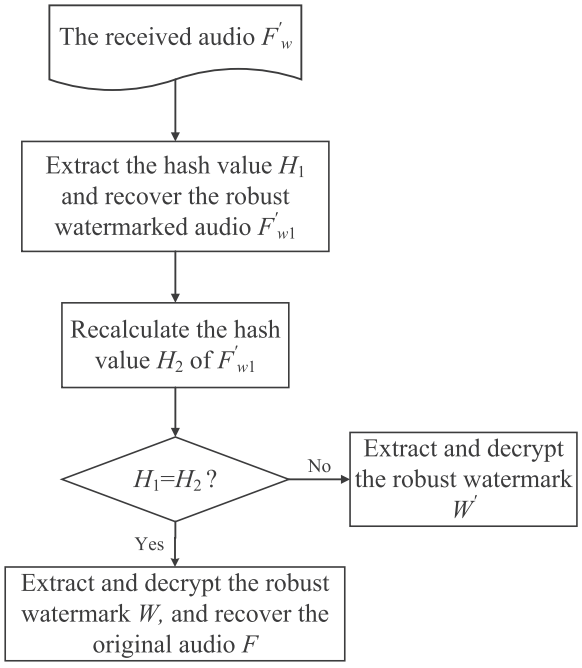


Fig. 3. The flowchart on integrity authentication in the transmission.

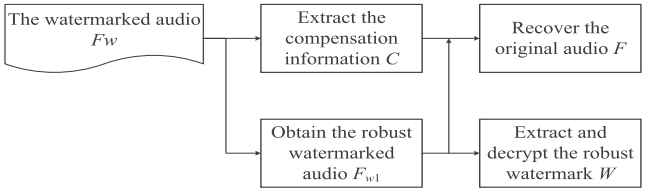


Fig. 4. Extract the watermark and recover the original audio without attacks.

3.7. Integrity authentication

In the process of transmission, the digital audio may be illegally tampered with by some malicious attackers. The proposed algorithm in this paper can not only protect digital audio copyright but also authenticate the integrity of digital audio.

As shown in Fig. 3, after receiving the watermarked audio F'_w, the receiver can extract the additional information by the exploited ReW way and then restore the robust watermarked audio F'_{w1} and the hash value H_1. Denote the hash value of F'_{w1} by H_2. If H_1 and H_2 are exactly the same, the watermarked audio has not been attacked in the process of transmission. In this case, the robust watermark can be further extracted, and the original audio can be recovered in a lossless way. Conversely, if H_1 is different from H_2, the watermarked audio has suffered some attacks during transmission. In this case, we can directly extract and decrypt the robust watermark W' from F'_w.

3.8. Extract the watermark and recover the original audio without attacks

If the watermarked audio is not attacked during transmission, the receiver can perfectly extract the robust watermark and recover the original audio without any distortion. The process of watermark extraction without any attacks is shown in Fig. 4.

a) *Extract the compensation information*: In Fig. 4, for a legal receiver, if the watermarked audio is not attacked during transmission, they can extract the watermark and recover the original audio in a lossless way. At first, the additional information is extracted with the secret key EN_1 and decompressed to get the compensation information and the hash value. Then, similar to the embedding algorithm, if the bit '1' appears in the decompressed robust distortion, it will be repeated three times to get four '1' to form the original robust distortion. As for an illegal receiver, it is difficult to reconstruct the original audio from the watermarked audio without the secret key EN_1.

b) *Extract the robust watermark*: From the extracted compensation information, we can get the bin width M, the length of the robust watermark L_w, the selected audio amplitude range B, the selected constant λ, and the robust distortion D. Then, the robust watermarked audio F_{w1} is obtained to compute the time-domain statistical histogram HS'. After that, in each bin each group of each four continuous samples in position will be organized to generate the histogram HS'_1. Since the shape of the low-frequency histogram is the same as that in the time domain, we can directly extract the robust watermark from the histogram HS'_1 in the time domain. How to extract the encrypted robust watermark is formulated as

$$\hat{w}(i) = \begin{cases} 1, & if\ \frac{a'}{b'} \geq 1 \\ 0, & else \end{cases} \tag{23}$$

After $\hat{W}$ is extracted, the encrypted sequence EN is calculated by using Eqs. (1) and (2). Finally, $\hat{W}$ can be decrypted as

$$w(i) = EN(i) \oplus \hat{w}(i) \tag{24}$$

According to Eqs. (23) and (24), the robust watermark information can be extracted without distortion in case of no attacks.

c) *Recover the original audio*: After obtaining the histogram HS'_1, if $a' > b'$ and the corresponding robust distortion is '1', for each group of two consecutive bins the original audio samples are recovered by

$$f'(i) = \begin{cases} f'(i) + M, & if\ a' > b' \\ f'(i), & else. \end{cases} \tag{25}$$

If $b' > a'$ and the corresponding robust distortion is '1', the recovery of the original audio is calculated as follows:

$$f'(i) = \begin{cases} f'(i) - M, & if\ b' > a' \\ f'(i), & else. \end{cases} \tag{26}$$

According to Eqs. (25) and (26), the original audio can be perfectly recovered in a lossless way.

3.9. Robust watermark extraction under the attacks

Digital watermarked audio may suffer from some malicious attacks (e.g., TSM, cropping, jitter, etc.) during transmission. Malicious attacks will cause permanent distortion for digital watermarked audio files. Therefore, in this case, the original audio cannot be recovered in a lossless way. However, although the original audio cannot be recovered under malicious attacks, the robust watermark can still be extracted from the attacked watermarked audio file $\hat{F}_w$ by computing its histogram and using two consecutive bins as a group to extract one bit, as shown in Eqs. (23) and (24).

4. Experimental results

In this section, we analyze the effectiveness of the proposed algorithm from the aspects of audio imperceptibility and robustness through experiments. The proposed algorithm can effectively authenticate the integrity and protect the copyright of digital audio. In the following experiments, signal-to-noise ratio (SNR) and perceptual evaluation of audio quality (PEAQ) are used to measure the imperceptibility of watermarked audio files objectively, and then a

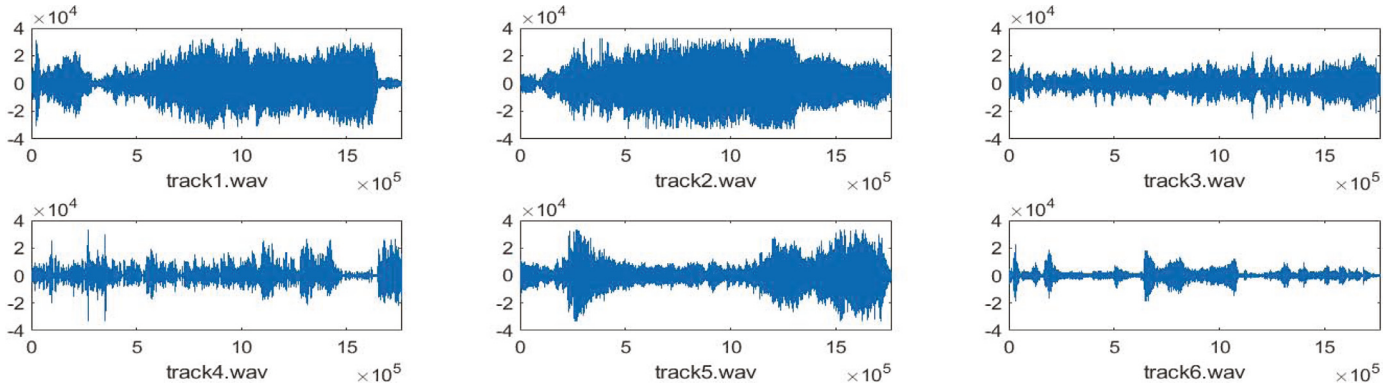


Fig. 5. Six typical audio segments.

Table 1
SNRs of the watermarked audio files and BER in case of no attacks.

Audio	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*
SNR(dB)	38.95	43.42	42.00	40.50	36.81	38.70
BER	0	0	0	0	0	0

subjective test is performed. Bit error ratio (BER) is used to measure the robustness of the proposed scheme. When the SNR of the watermarked audio is greater than 20 dB, it can be considered that the algorithm has achieved better imperceptibility. Besides, if the BER is less than 20%, it can be considered that the robust watermark can effectively protect the copyright of the digital audio. In our experiments, the parameter λ is 3.6, and the threshold T is 6. The experimental results show that the algorithm proposed in this paper can realize acceptable imperceptibility and strong robustness. Furthermore, in case of attacks, it can resist some additive noise-like operations and desynchronization attacks, such as TSM, cropping, and jitter.

4.1. Objective test of watermark imperceptibility

In this subsection, the imperceptibility of the watermarked audio files is measured objectively by using the SNR value and PEAQ, respectively. In the imperceptibility aspect, the embedding distortion mainly comes from the insertion of the robust watermark, the compensation information, and the hash value. Therefore, six audio files (*track1-track6*) from https://sound.media.mit.edu/media.php are selected for testing the imperceptibility. In this paper, the selected audio files are cropped into around 40s (44.1kHz, mono) audio segments, as shown in Fig. 5.

4.1.1. Watermark imperceptibility measured by SNR

The SNR values of the marked audio files are listed in Table 1. It can be seen from Table 1 that after the 50-bit (1.25 bits per second (bps)) robust watermark information is embedded, the SNR values of the six watermarked audio segments are over 35 dB, indicating that the acceptable imperceptibility has been achieved. At the same time, in case of no attacks, the receiver can extract the robust watermark information accurately and recover the original audio segments in a lossless way.

Take the *track1* audio segment as an example. In Fig. 6, Fig. 6(a) represents the robust watermarked audio segment with SNR of 39.09 dB, Fig. 6(b) is the watermarked audio segment after embedding the additional information for reversibility (SNR=38.95 dB), and Fig. 6(c) is the difference between the original and recovered segments in case of no attacks. It can be seen from Fig. 6 that the algorithm proposed in this paper not only has good imperceptibility but also can restore the original audio segment without any distortion in case of no attacks.

Table 2
Objective difference grade.

ODG	Description of impairments	Quality
0.0	Imperceptible	Excellent
-1.0	Perceptible, but not annoying	Good
-2.0	Slightly annoying	Fair
-3.0	Annoying	Poor
-4.0	Very annoying	Bad

Table 3
ODG values of the watermarked audio files.

Audio	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*
ODG	-3.60	-1.45	-2.36	-1.94	-3.89	-3.76

In addition, in order to further investigate the impact of the additional information embedding on the imperceptibility of digital audio, we randomly selected 120 audio segments (including classical, folk, jazz, pop, rock, and spoken) from the online site: https://freemusicarchive.org/home, which are further cut into around 40s (44.1 kHz, mono) audio segments for embedding a 50-bit (1.25 bps) robust watermark sequence in each segment. The experimental results show that when only the robust watermark is embedded, the average SNR value of the robust watermarked audio is 39.15 dB while the average SNR value of the final watermarked audio is 38.72 dB after embedding the additional information for reversibility. We can see from these experimental results that the way to embed the additional information for the reversibility has an effect on the average value of SNR, which makes it decrease by 0.43 dB. Since the average SNR value of the final marked audio segments is far greater than 20 dB, we consider that the method has an acceptable imperceptibility.

4.1.2. Watermark imperceptibility measured by PEAQ

In addition to using SNR values to evaluate the imperceptibility of the watermarked audio, we also perform the PEAQ on watermarked audio by using EAQUAL software. As shown in Table 2, the measurement results are expressed using the objective difference grade (ODG) standard, which has five levels to measure the imperceptibility of the audio [59]. The closer the ODG value is to zero, the better the imperceptibility of the watermarked audio. Table 3 lists the PEAQ test results, the ODG values of which are between -1.45 and -3.89, indicating that the watermark distortion is acceptable from the angle of the PEAQ measurement.

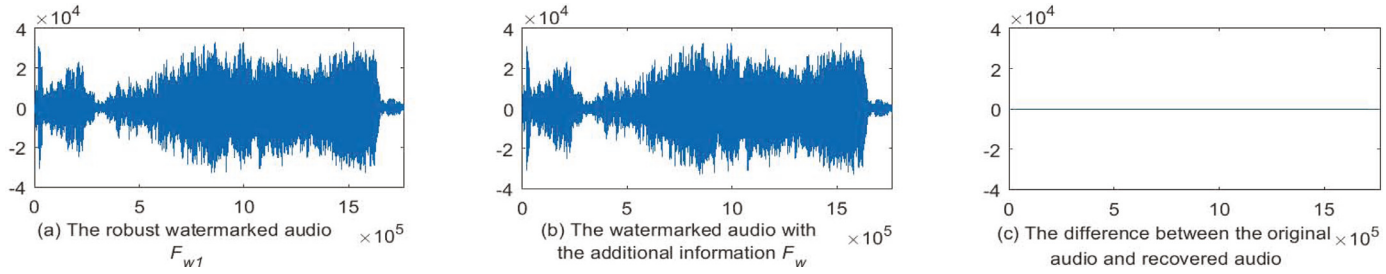


Fig. 6. (a) *Track1* after embedding the robust watermark (39.09 dB). (b) *Track1* after embedding the robust watermark and the additional information (38.95 dB). (c) The difference between the original and recovered audio segments.

Table 4
Results of PEAQ by 50 testers.

Description	Number of testers
Imperceptible	5
Perceptible, but not annoying	16
Slightly annoying	29
Annoying	0
Very annoying	0

Table 5
Robustness performance to common additive noise-like operations.

Attack	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*
Gau(45dB)	0/50	0/50	0/50	0/50	0/50	0/50
Gau(40dB)	0/50	0/50	0/50	0/50	0/50	0/50
Gau(35dB)	0/50	0/50	0/50	0/50	0/50	0/50
LP(8kHz)	0/50	2/50	0/50	1/50	0/50	0/50
LP(7kHz)	0/50	3/50	0/50	0/50	0/50	0/50
LP(6kHz)	0/50	6/50	1/50	0/50	0/50	0/50

Table 6
Robustness performance to some common attacks in StirMark benchmark for audio.

Attack	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*
addbrumm_100	0/50	0/50	1/50	0/50	0/50	8/50
amplify	0/50	0/50	0/50	0/50	0/50	4/50
addnoise_100	0/50	0/50	0/50	0/50	0/50	0/50
compressor	3/50	3/50	0/50	0/50	0/50	0/50
cutsample	0/50	0/50	0/50	0/50	0/50	0/50
extrastereo_30	0/50	0/50	0/50	0/50	0/50	0/50
extrastereo_50	0/50	0/50	0/50	0/50	0/50	0/50
extrastereo_70	0/50	0/50	0/50	0/50	0/50	0/50
exchange	0/50	0/50	0/50	0/50	0/50	0/50
FFT_realreverse	0/50	0/50	0/50	0/50	0/50	0/50
flippsample	0/50	0/50	0/50	0/50	0/50	0/50
normalize	0/50	0/50	0/50	0/50	0/50	0/50
nothing	0/50	0/50	0/50	0/50	0/50	0/50
original	0/50	0/50	0/50	0/50	0/50	0/50
rc_lowpass	0/50	0/50	0/50	0/50	0/50	0/50
stat1	0/50	0/50	0/50	0/50	0/50	0/50
stat2	0/50	0/50	0/50	0/50	0/50	0/50
smooth	0/50	0/50	0/50	0/50	0/50	0/50
smooth2	0/50	0/50	0/50	0/50	0/50	0/50
zerocross	0/50	0/50	2/50	3/50	1/50	-
zerolength	0/50	0/50	0/50	0/50	0/50	0/50
zeroremove	0/50	0/50	0/50	0/50	0/50	0/50

4.2. Subjective test of watermark imperceptibility

In order to test the imperceptibility more realistically, ten audio segments (including original and watermarked audio) are randomly selected, and 50 testers are randomly invited for the subjective tests. In the experiments, each tester is required to discriminate whether the audio segment heard contains watermark information. The results of 50 testers are listed in Table 4.

Table 7
Robustness performance to MP3 lossy compression.

Attack	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*
128(kbps)	0/50	0/50	1/50	0/50	0/50	0/50
112(kbps)	0/50	1/50	0/50	0/50	0/50	0/50
96(kbps)	0/50	5/50	1/50	1/50	0/50	2/50
80(kbps)	0/50	6/50	1/50	1/50	0/50	3/50
64(kbps)	0/50	8/50	3/50	1/50	0/50	3/50
56(kbps)	1/50	-	5/50	2/50	0/50	6/50
48(kbps)	5/50	-	9/50	3/50	0/50	8/50

Table 8
Robustness performance to the pitch-invariant TSM attacks.

Pitch Invariance	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*
TSM -20%	0/50	0/50	0/50	0/50	0/50	0/50
TSM -15%	0/50	0/50	0/50	0/50	0/50	0/50
TSM -10%	0/50	0/50	0/50	0/50	0/50	0/50
TSM -5%	0/50	0/50	0/50	0/50	0/50	0/50
TSM 5%	0/50	0/50	0/50	0/50	0/50	0/50
TSM 10%	0/50	0/50	0/50	0/50	0/50	0/50
TSM 15%	0/50	0/50	0/50	0/50	0/50	0/50
TSM 20%	0/50	0/50	0/50	0/50	0/50	0/50

Table 9
Robustness performance to the resample TSM attacks.

Resample Invariance	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*
TSM -20%	0/50	0/50	0/50	0/50	0/50	0/50
TSM -15%	0/50	0/50	0/50	0/50	0/50	0/50
TSM -10%	0/50	0/50	0/50	0/50	0/50	0/50
TSM -5%	0/50	0/50	0/50	0/50	0/50	0/50
TSM 5%	0/50	0/50	0/50	0/50	0/50	0/50
TSM 10%	0/50	0/50	0/50	0/50	0/50	0/50
TSM 15%	0/50	0/50	0/50	0/50	0/50	0/50
TSM 20%	0/50	0/50	0/50	0/50	0/50	0/50

It can be seen from Table 4 that in the experiment of the 50 testers, 10% (5 testers) of testers cannot distinguish between the original audio and watermarked audio files. And, 32% (16 testers) of testers are perceptible, but the auditory perception is not annoying. 58% (29 testers) of testers are perceptible, and the auditory perception is slightly annoying. In general, according to the experimental results in Table 4, most of the testers considered that the watermarked audio segments using the ALRW scheme proposed in this article are slightly annoying. Therefore, we consider that the algorithm introduced in this paper has an acceptable imperceptibility.

4.3. Robustness testing

In the process of digital audio transmission, the marked audio files may suffer from some additive noise-like operations (such as

Table 10
Robustness performance to the TIPS attacks.

Scaling factors	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*
TIPS 80%	0/50	0/50	0/50	0/50	0/50	0/50
TIPS 85%	0/50	0/50	0/50	0/50	0/50	0/50
TIPS 90%	0/50	0/50	0/50	0/50	0/50	0/50
TIPS 95%	0/50	0/50	0/50	0/50	0/50	0/50
TIPS 105%	0/50	0/50	0/50	0/50	0/50	0/50
TIPS 110%	0/50	0/50	0/50	0/50	0/50	0/50
TIPS 115%	0/50	0/50	0/50	1/50	0/50	0/50
TIPS 120%	0/50	2/50	2/50	0/50	0/50	0/50

Table 11
Robustness performance to jitter and cropping attacks.

Attack	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*
Jitter(1/10)	0/50	0/50	0/50	0/50	0/50	0/50
Jitter(1/50)	0/50	0/50	0/50	0/50	0/50	0/50
Jitter(1/100)	0/50	0/50	0/50	0/50	0/50	0/50
Jitter(1/300)	0/50	0/50	0/50	0/50	0/50	0/50
Jitter(1/500)	0/50	0/50	0/50	0/50	0/50	0/50
Crop(5%)	0/50	0/50	0/50	0/50	0/50	0/50
Crop(10%)	0/50	0/50	0/50	0/50	0/50	0/50
Crop(15%)	0/50	0/50	0/50	0/50	0/50	0/50

filtering, additive noises, MP3 compression, etc.) and desynchronization attacks (such as TSM, cropping, jitter, etc.). After attacks, we hope that the robust watermark can be still detected for audio copyright protection. In this paper, the six audio segments plotted in Fig. 5 are selected to test the robustness of the proposed ALRW algorithm by embedding a 50-bit (1.25 bps) robust watermark sequence, and the BER values for different attacks are listed in Table 5 to Table 11.

In Table 5, we use the GoldWave software to perform a low-pass filtering (LP) operation on the watermarked audio. It can be seen that the BER values of the six audio segments are less than 20%, meaning the proposed algorithm can resist the additive white Gaussian (Gau) noise and low-pass filtering operations. After the attacks with the StirMark Benchmark for Audio, Table 6 shows the proposed algorithm can resist more than 18 kinds of StirMark attacks, such as *addbrumm100, amplify, addnoise100, etc*. Table 7 lists the experimental results on MP3 compression operations. For MP3 compression from 128 kbps to 48 kbps on the watermark, the experimental results show that the watermarked audio segment *track2* can resist up to 64 kbps while the other five tracks can resist the MP3 compression of 48 kbps. These results show that the proposed ALRW algorithm is robust for common additive noise-like manipulations.

For desynchronization attacks, we use the CoolEditPro software to perform two types of TSM operations on the watermarked audio files. As listed in Table 8 and Table 9, for the six tested audio tracks, the BER values under the two types of TSM attacks (the pitch-invariant and resample TSM modes) are 0, indicating that the proposed watermarking scheme has perfect robustness to these two kinds of TSM attacks with different scaling factors. Furthermore, the time-invariant pitch scaling (TIPS) attacks are performed in case of the scaling factors from 80% to 120%. The experimental results are listed in Table 10, and we can easily observe that the ALRW algorithm proposed in this paper is insensitive to these TIPS attacks. Finally, we test jitter and cropping attacks, and the experimental results are listed in Table 11. For jitter 1/10 to 1/500, the robust watermark can be successfully extracted from all selected audio segments while the watermark is robust for random cropping of 5%, 10%, and 15%. In other words, these results show the proposed ALRW algorithm provides satisfactory robustness to those challenging desynchronization attacks.

4.4. Performance comparison

In this part, we compared the proposed algorithm against two existing histogram-based RoW algorithms [21,22] to show the resistance of the histogram shape to various attacks. The experimental results for different attacks have been listed in Table 12 to Table 16, respectively. In these five tables, six audio segments (from *track1* to *track6*) are selected for robustness testing by inserting a 40-bit (1 bps) robust watermark sequence.

We can see from Table 12 that in the proposed ALRW algorithm, the SNR values of the six audio clips are all over 30 dB, which achieves an acceptable imperceptibility. In fact, the reversible embedding of the additional information into the robust watermarked audio has an effect on the imperceptibility, where the SNR values of the final marked audio are a little lower than that of the robust watermarked audio. Take *track*1 as an example, the SNR value of the robust watermarked audio is 36.61 dB, 0.09 dB higher than that of the final watermarked audio. In the methods [21,22], the SNR values of the watermarked audio files are higher than 42 dB due to the fact that there is no additional information to be embedded for the reversibility. However, the SNR values of the proposed algorithm are still higher than 30 dB, satisfying the imperceptibility requirement while it has an ability to recover the cover signal.

In Tables 13 and 14, we compare the robustness of these three algorithms to the pitch-invariant TSM and the resample TSM operations, respectively. We can see from these two tables that the BER values of the proposed algorithm and the method [22] for the six test files are all Zero due to the fact that the robust watermark information is embedded in the DWT domain. As for [21], a few of the watermark bits can not be extracted correctly under resample TSM attacks due to the watermark inserted in the time domain.

Table 15 reports the robustness to the MP3 compression attacks under the different compression ratios. We can see from this table that the ALRW algorithm proposed in this paper can resist MP3 compression of 56 kbps for all audio segments after embedding the 40-bit (1 bps) robust watermark. Compared with the RoW schemes in [21] and [22], the proposed one is a little bit better.

Table 16 shows the comparison results under cropping and jitter attacks. It can be seen from Table 16 that all three algorithms can resist cropping and jitter attacks well. For example, in the case of cropping 5%, 10%, 15%, and jitter 1/10, 1/50, 1/100, 1/300, 1/500, the BER values of the watermark bits are Zero.

4.5. Compared with previous RoW methods

In this part, we made a detailed comparison with other previous RoW schemes. Table 17 lists the robustness and reversibility comparison of the proposed algorithm with other robust audio watermarking methods. From Table 17, it can be seen that in terms of pitch-invariance TSM attacks, the algorithm proposed in [22] is extremely insensitive due to the fact that the robust watermark information is embedded into the low-frequency component of DWT to obtain stronger robustness. Although the proposed method in this paper also embeds the robust watermark in the low-frequency component of DWT, the proposed algorithm additionally embeds the robust distortion, which has an impact on robustness. As for resample TSM attacks, except for [19], the robust watermarking algorithms in Table 17, including the scheme presented in this paper, are resistant to these attacks with scaling factors from 80% to 120%. For MP3 compression, the schemes presented in [19] and [28] can resist MP3 compression attacks at 32 kbps, while the proposed one is robust for the compression ratio of 48 kbps. In terms of jitter attacks, the method proposed in [19] is robust for the factors from 1/500 to 1/1000, and the other RoW methods in Table 17 can resist jitter attacks with the factors from 1/10 to 1/500. In addition, the proposed method has reversibility in case of no attacks, and

Table 12
Comparison of SNR and BER.

	Proposed ALRW algorithm						Xiang *et al.* [21]						Xiang *et al.* [22]					
	track1	*track2*	*track3*	*track4*	*track5*	*track6*	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*
SNR(dB)	36.52	40.78	39.15	38.60	34.32	36.27	42.73	42.69	42.50	44.21	43.53	44.33	42.59	42.59	42.49	43.93	43.28	44.21
BER	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 13
Comparison of robustness performance to the pitch-invariant TSM.

Pitch-invariant TSM	Proposed ALRW algorithm						Xiang *et al.* [21]						Xiang *et al.* [22]					
	track1	*track2*	*track3*	*track4*	*track5*	*track6*	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*
TSM -20%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40
TSM -15%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40
TSM -10%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40
TSM -5%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40
TSM 5%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40
TSM 10%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40
TSM 15%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40
TSM 20%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40

Table 14
Comparison of robustness performance to the resample TSM.

Resample TSM	Proposed ALRW algorithm						Xiang *et al.* [21]						Xiang *et al.* [22]					
	track1	*track2*	*track3*	*track4*	*track5*	*track6*	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*
TSM -20%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	6/40	2/40	1/40	3/40	0/40	0/40	0/40	0/40	0/40	0/40
TSM -15%	0/40	0/40	0/40	0/40	0/40	0/40	1/40	4/40	5/40	2/40	0/40	2/40	0/40	0/40	0/40	0/40	0/40	0/40
TSM -10%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	1/40	5/40	1/40	1/40	3/40	0/40	0/40	0/40	0/40	0/40	0/40
TSM -5%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	3/40	3/40	2/40	0/40	3/40	0/40	0/40	0/40	0/40	0/40	0/40
TSM 5%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	4/40	4/40	1/40	0/40	3/40	0/40	0/40	0/40	0/40	0/40	0/40
TSM 10%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	1/40	5/40	2/40	0/40	4/40	0/40	0/40	0/40	0/40	0/40	0/40
TSM 15%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	1/40	3/40	1/40	0/40	3/40	0/40	0/40	0/40	0/40	0/40	0/40
TSM 20%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	4/40	2/40	0/40	2/40	0/40	0/40	0/40	0/40	0/40	0/40

Table 15
Comparison of robustness performance to MP3 compression.

Attack	Proposed ALRW algorithm						Xiang *et al.* [21]						Xiang *et al.* [22]					
	track1	*track2*	*track3*	*track4*	*track5*	*track6*	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*
128kbps	0/40	0/40	0/40	0/40	0/40	0/40	0/40	1/40	0/40	0/40	0/40	1/40	0/40	0/40	0/40	1/40	0/40	2/40
112kbps	0/40	0/40	0/40	0/40	0/40	0/40	0/40	4/40	2/40	0/40	0/40	1/40	0/40	0/40	0/40	0/40	0/40	2/40
96kbps	0/40	0/40	0/40	0/40	0/40	0/40	0/40	5/40	2/40	2/40	1/40	3/40	0/40	2/40	1/40	0/40	1/40	2/40
80kbps	0/40	3/40	0/40	0/40	0/40	0/40	0/40	-	3/40	1/40	1/40	5/40	0/40	5/40	1/40	4/40	1/40	7/40
64kbps	0/40	4/40	0/40	1/40	0/40	0/40	1/40	-	6/40	6/40	2/40	4/40	3/40	5/40	3/40	6/40	3/40	6/40
56kbps	0/40	6/40	1/40	0/40	0/40	0/40	2/40	-	-	6/40	1/40	7/40	3/40	-	-	-	-	7/40
48kbps	0/40	-	4/40	1/40	0/40	1/40	5/40	7/40	-	7/40	4/40	-	6/40	-	-	-	6/40	5/40

Table 16
Comparison of robustness performance to cropping and jitter attacks.

Attack	Proposed ALRW algorithm						Xiang *et al.* [21]						Xiang *et al.* [22]					
	track1	*track2*	*track3*	*track4*	*track5*	*track6*	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*	*track1*	*track2*	*track3*	*track4*	*track5*	*track6*
Crop -5%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40
Crop -10%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40
Crop -15%	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40
Jitter(1/10)	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40
Jitter(1/50)	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40
Jitter(1/100)	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40
Jitter(1/300)	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40
Jitter(1/500)	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40	0/40

Table 17
Comparison of the proposed one against previous RoW algorithms.

Algorithm	Pitch-invariant TSM	Resample TSM	MP3 compression	Jitter	Reversibility
In [19]	Scaling factors from 83% - 116%	Not reported	32 kbps	1/500 - 1/1000	No
In [21]	Scaling factors from 80% - 130%	Scaling factors from 70% - 130%	128 kbps	1/10 - 1/500	No
In [22]	Scaling factors from 70% - 130%	Scaling factors from 80% - 120%	64 kbps	1/10 - 1/500	No
In [28]	Scaling factors from 80% - 120%	Scaling factors from 75% - 140%	32 kbps	1/10 - 1/1500	No
In [38]	Scaling factors from 80% - 120%	Scaling factors from 80% - 120%	64 kbps	1/10 - 1/1000	No
Proposed method	Scaling factors from 80% - 120%	Scaling factors from 80% - 120%	48 kbps	1/10 - 1/1000	Yes

Table 18
Comparison of the proposed one against three previous ALRW algorithms.

Audio	Proposed ALRW algorithm				Xiang *et al.* [41]				Liang *et al.* [43]				Nishimura [55]			
	ODG	Capacity (bit)	MP3 (80kbps)	Gau (36dB)	ODG	Capacity (bit)	MP3 (80kbps)	Gau (36dB)	ODG	Capacity (bit)	MP3 (64kbps)	Gau (36dB)	ODG	Capacity (bit)	MP3 (128kbps)	Gau (36dB)
track65	-1.91	50	6%	0	-0.23	1000	0	1.2%	-0.21	1000	0	0.54%	-1.21	216	0	6%
track66	-1.25	50	-	6%	-1.05	1000	0.3%	0	-1.03	1000	0.7%	0	-0.24	216	0	6%
track69	-0.95	50	14%	4%	-0.23	1000	0	0	-0.22	1000	0	0	-0.31	216	1%	4%

the other robust audio methods cannot restore the original audio signal due to the permanent distortion caused by the watermark embedding process.

4.6. Compared with previous ALRW algorithms

In the literature, researchers have presented some ALRW schemes for common additive noise-like operations, but there are currently no reports on ALRW algorithms against desynchronization attacks. For a clear comparison, in Table 18, three audio clips (*track65, track66*, and *track69*) with a sample rate of 44.1kHz are selected from [60] for the performance comparison of the proposed method against several typical ALRW algorithms.

It can be seen from Table 18 that the ODG values of the algorithm proposed in this paper are between -0.95 and -1.91 since the robust watermark information is embedded in the low-frequency component of DWT, while the other three methods are between -0.21 and -1.21. In addition, the algorithms proposed in [41], [43], and [55] can provide a larger embedding capacity than the proposed one. The basic reason is that the previous methods in [41] and [43] embed bits by using the histogram shifting technique, which can insert more bits by shifting the Zero bin. In the proposed scheme, the capacity is about 50 bits (1.25 bps) due to the limitation of the bins in number under the constraint of robustness and imperceptibility. In terms of robustness to MP3 compression, except for *track66*, the watermark in the other two tracks in Table 18 is robust to the compression rate of 80 kbps by using the proposed scheme, while the methods in [41], [43], and [55] can resist the compression rate of 80 kbps, 64 kbps, and 128 kbps, respectively. As for additive Gaussian noise, all algorithms in Table 18 are robust for the power of 36 dB. Of course, the proposed algorithm in this paper provides a new ability to desynchronization attacks (e.g., TSM, cropping, jitter, etc.), while the prior ALRW methods proposed in [41], [43], [54], and [55] are sensitive to desynchronization attacks since the watermark in those prior ALRW algorithms is related to the spatial or time position.

In summary, the ALRW algorithm presented in this paper has an acceptable imperceptibility while embedding 50 bits of the watermark information. In case of attacks, the proposed ALRW method can resist some common additive noise-like operations and overcome several challenging desynchronization attacks by marking the histogram shape. And in case of no attacks, the robust watermark can be extracted without any distortion, and the original audio signal can be recovered in a lossless way due to its reversibility.

5. Conclusion

This paper presents a new ALRW algorithm that is not only resistant to common additive noise-like operations but also robust for desynchronization attacks. In case of no attacks, the original audio can be recovered without any loss. By testing the 120 audio segments, we have shown that the average SNR value of the watermarked audio file is 38.72 dB, and the watermark is resistant to various attacks, indicating that the proposed scheme has good imperceptibility and robustness.

The proposed ALRW scheme has two highlights. One is that this is the first report on the ALRW algorithm against desynchronization attacks. In the previous ALRW algorithms, the watermark is resistant to common additive noise-like operations, but very fragile to those challenging desynchronization attacks like TSM, cropping a part of the cover audio, and jitter attacks. More importantly, compared with those traditional RoW algorithms, the proposed one can recover the original audio in case of no attacks while keeping satisfactory robustness. In such a way, it can avoid permanent distortion due to the robust watermark embedding. Besides, the proposed method can effectively verify whether the digital audio signal has been illegally tampered with by malicious attackers in the process of transmission by using the hash value of the robust watermarked audio as a part of the additional information. Extensive results have shown that the proposed ALRW algorithm is reversible and provides satisfactory robustness to additive noise-like manipulations and desynchronization attacks, including MP3 compression with compression rates from 128 kbps to 64 kbps, TSM with the factors from -20% to 20%, random deleting one in each group of 10 samples, and random cropping with the factors from 5% to 15%.

In future works, how to improve the imperceptibility and capacity of the robust watermark is one of our considerations. Besides, more comparison works on the proposed method against the previous techniques will be another future consideration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Jingxuan Li: Conceptualization, Methodology, Software, Data curation, Writing – original draft. **Shijun Xiang:** Conceptualization, Methodology, Validation.

Acknowledgemnts

This work was supported by the National Natural Science Foundation of China (No. 61772234).

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